Integration Testing Report

Hostel Management System — Quaid-i-Azam University

1. Introduction

Integration testing is a software testing technique used to verify that different modules of an application work correctly when connected. It focuses on communication between modules, data transfer, and consistency of information throughout the system.

The Hostel Management System is designed to manage student applications, hostel rooms, bed allotments, fee payments, complaints, visitors, notices, and reports. Since these functions are related, an error in one module may affect the operation of another module.

The purpose of this report is to define integration test cases for the Hostel Management System and provide a structured method for recording actual test results.

2. Objectives

The main objectives of integration testing are:

- Verify communication between connected modules.
- Ensure that data entered in one module is correctly available in related modules.
- Verify that application approval is connected with bed allotment.
- Ensure that room occupancy is updated correctly after allotment.
- Verify that complaint status updates are reflected in relevant staff views.
- Ensure that payment verification is reflected in the student's fee status.
- Verify that visitor records are associated with the correct hostel resident.
- Identify integration defects and maintain data consistency.

3. Scope of Testing

The integration testing process covers the following module interactions:

1. Authentication and role-based access.
2. Student profile and hostel application.
3. Application review and application tracking.
4. Application approval and bed allotment.
5. Bed allotment and room occupancy.
6. Complaint submission and complaint-status management.

7. Payment verification and student fee status.
8. Visitor entry and resident records.
9. Notice management and student access.
10. Operational records and report generation.

4. Testing Environment

The following information should be completed according to the actual environment used for testing.

Item	Details
Project	Hostel Management System
Institution	Quaid-i-Azam University
Testing type	Integration Testing
Test environment	To be recorded
Database	To be recorded
Application version	To be recorded
Testing date	To be recorded
Tester	To be recorded

5. Integration Test Cases

The following test cases are designed according to the requirements and workflows described in the project documentation. The actual results and final status must be recorded after execution.

IT-01: Student Profile and Hostel Application

Objective: Verify that the hostel application is associated with the correct student profile.

Preconditions: A valid student account exists, and the student can access the application form.

Test steps:

1. Log in using a valid student account.
2. Open the hostel application form.

3. Enter the required application information.
4. Submit the application.
5. Check the application record and student information.

Expected result: The application should be saved successfully and associated with the correct student. The application should be available for staff review.

Actual result: To be recorded after execution.

Status: Not Run.

IT-02: Application Approval and Bed Allotment

Objective: Verify the connection between application review and bed allotment.

Preconditions: A pending application exists, and an authorized staff account is available.

Test steps:

1. Log in as authorized hostel staff.
2. Open a pending hostel application.
3. Approve the application.
4. Open the bed-allotment module.
5. Locate the approved student's record.
6. Attempt to assign an available bed.

Expected result: The approved application should be available for the allotment process. The system should associate the allotment with the correct student and prevent allocation of an unavailable bed.

Actual result: To be recorded after execution.

Status: Not Run.

IT-03: Bed Allotment and Room Occupancy

Objective: Verify that bed allotment updates room occupancy correctly.

Preconditions: A room exists with available capacity, and a student is eligible for allotment.

Test steps:

1. Record the room's current occupancy and capacity.

2. Assign an available bed to the student.
3. Open the room-management module.
4. Check the updated occupancy information.
5. Attempt to allot the same bed to another student.

Expected result: Room occupancy should reflect the successful allotment. The same bed should not be assigned to two students, and the room capacity should not be exceeded.

Actual result: To be recorded after execution.

Status: Not Run.

IT-04: Complaint Submission and Staff Dashboard

Objective: Verify that a complaint submitted by a student becomes available to authorized staff.

Preconditions: A valid student account and an accessible complaint-submission function are available.

Test steps:

1. Log in as a student.
2. Submit a complaint with the required details.
3. Open the complaint-management view using an authorized staff account.
4. Locate the submitted complaint.
5. Update its status.
6. Check the updated status in the relevant student view.

Expected result: The complaint should be associated with the correct student and appear in the appropriate staff view. Any status update should be reflected in the relevant view.

Actual result: To be recorded after execution.

Status: Not Run.

IT-05: Payment Verification and Student Fee Status

Objective: Verify that payment verification is reflected in the student's fee information.

Preconditions: A student fee record and an authorized staff account are available.

Test steps:

1. Open the student's fee or payment record.
2. Enter or locate a payment reference and the required payment details.
3. Verify the payment using the available verification process.
4. Open the student's fee-status view.
5. Compare the displayed status with the verification record.

Expected result: The fee status should reflect the verified payment correctly. An unverified payment should not be incorrectly displayed as verified.

Actual result: To be recorded after execution.

Status: Not Run.

IT-06: Visitor Entry and Resident Record

Objective: Verify that visitor records are associated with the correct hostel resident.

Preconditions: A valid resident record is available, and the appropriate visitor-management access is provided.

Test steps:

1. Open the visitor-management module.
2. Enter the visitor's required information.
3. Select the correct resident and record the visit.
4. Check the saved visitor entry.
5. Record the visitor's exit when applicable.
6. Review the updated visitor record.

Expected result: The visitor entry should contain the correct resident and visit details. The exit information should be associated with the correct entry.

Actual result: To be recorded after execution.

Status: Not Run.

IT-07: Notice Management and Student Access

Objective: Verify that published notices are available to the intended users.

Preconditions: An authorized staff account and a student account are available.

Test steps:

1. Log in as authorized staff.

2. Create and publish a notice.
3. Open the student-facing notice view.
4. Check whether the published notice appears.
5. Verify that any available notice information is displayed correctly.

Expected result: The published notice should appear in the appropriate student view with correct information.

Actual result: To be recorded after execution.

Status: Not Run.

IT-08: Data Records and Report Generation

Objective: Verify that reports use correct information from the related modules.

Preconditions: Relevant application, allotment, payment, or visitor records exist, and the user has report-generation permission.

Test steps:

1. Create or identify sample records in the relevant modules.
2. Open the report-generation module.
3. Select the appropriate report type and filters.
4. Generate the report.
5. Compare its contents with the corresponding records.

Expected result: The report should contain accurate information for the selected filters, without missing, duplicated, or incorrectly associated records.

Actual result: To be recorded after execution.

Status: Not Run.

6. Test Execution Summary

The following table should be updated after the test cases have been executed.

Test ID	Integration Area	Status
IT-01	Student profile and application	Not Run
IT-02	Application approval and allotment	Not Run
IT-03	Allotment and occupancy	Not Run

IT-04	Complaint and staff dashboard	Not Run
IT-05	Payment verification and fee status	Not Run
IT-06	Visitor and resident records	Not Run
IT-07	Notice management and student access	Not Run
IT-08	Records and report generation	Not Run

Summary: Eight integration test cases have been defined for this report. The number of passed and failed tests will be determined after execution. No conclusion about successful integration should be made until actual results have been recorded.

7. Defect Reporting and Handling

If an integration test fails, the defect should be documented with the following information:

- Defect ID.
- Related test case ID.
- Module or modules affected.
- Steps to reproduce the problem.
- Expected result.
- Actual result.
- Severity and priority.
- Corrective action taken.
- Retest result and final status.

After a defect is corrected, the affected test should be repeated. Related integration tests should also be performed to check whether the correction has introduced another problem.

8. Acceptance Criteria

The integration testing process will be considered satisfactory when:

1. Connected modules exchange and display the correct information.
2. Approved applications can proceed through the intended allotment workflow.
3. Room occupancy remains consistent with bed-allotment records.

4. Complaint and payment status information remains consistent across relevant views.
5. Visitor entries and exits are associated with the correct records.
6. Reports accurately represent the underlying data.
7. Critical integration defects have been resolved and retested.

9. Conclusion

Integration testing is an important part of validating the Hostel Management System because its modules depend on shared records and connected workflows. This report defines test cases for the major interactions between student applications, room and bed allotment, complaint management, fee verification, visitor management, notices, and reporting.

Executing these test cases and recording the actual outcomes will help identify integration defects, verify data consistency, and assess whether the connected modules satisfy the system requirements. The final testing conclusion should be prepared after reviewing the recorded results and resolving any significant defects.